

РБНФ №1 (опис синтаксису мови всіма допустимими засобами РБНФ)	РБНФ №2 (опис формальна граматика мови)	Опис формальної граматики мови	Опис граматики з специфікацією lookahead для LL2-аналізатора	/* Код для перевірки РБНФ №1 */	/* Код для перевірки РБНФ №2 */	/* Дані для LL2-аналізатора (лексми вже опрацьовані лексичним аналізатором) УВАГА: при копіюванні зважайте, щоб у кожному рядку після символу «\» не містилось жодних інших символів. */
		G = (N, T, P, S)	G = (N, T, P, S)			
		S → program_rule	S → program_rule			
		N = { labeled_point, goto_label, program_name, value_type, array_specify, declaration_element, array_specify__optional, other_declaration_ident, declaration, other_declaration_ident__iteration, index_action, unary_operator, unary_operation, binary_operator, binary_action, left_expression, group_expression, index_action__optional, expression, binary_action__iteration, expression_or_cond_block__with_optional_assign, assign_to_right, assign_to_right__optional, if_expression, body_for_true, false_cond_block_without_else, body_for_false, cond_block, false_cond_block_without_else__iteration, body_for_false__optional, statement, block_statements, input_rule, argument_for_input, output_rule, statement__iteration, expression__optional, program_rule, declaration__optional, non_zero_digit, digit__iteration, digit, unsigned_value, value, sign__optional, sign, ident, letter_in_upper_case, letter_in_lower_case, sign_plus, sign_minus }	N = { labeled_point, goto_label, program_name, value_type, array_specify, declaration_element, array_specify__optional, other_declaration_ident, declaration, other_declaration_ident__iteration, index_action, unary_operator, unary_operation, binary_operator, binary_action, left_expression, group_expression, index_action__optional, expression, binary_action__iteration, expression_or_cond_block__with_optional_assign, assign_to_right, assign_to_right__optional, if_expression, body_for_true, false_cond_block_without_else, body_for_false, cond_block, false_cond_block_without_else__iteration, body_for_false__optional, statement, block_statements, input_rule, argument_for_input, output_rule, statement__iteration, expression__optional, program_rule, declaration__optional, non_zero_digit, digit__iteration, digit, unsigned_value, value, sign__optional, sign, ident, letter_in_upper_case, letter_in_lower_case, sign_plus, sign_minus }	#define NONTERMINALS labeled_point, \ goto_label, \ program_name, \ value_type, \ array_specify, \ declaration_element, \ \ other_declaration_ident, \ declaration, \ \ index_action, \ unary_operator, \ unary_operation, \ binary_operator, \ binary_action, \ left_expression, \ group_expression, \ \ expression, \ \ expression_or_cond_block__with_optional_assign, \ assign_to_right, \ \ if_expression, \ body_for_true, \ false_cond_block_without_else, \ body_for_false, \ cond_block, \ \ statement, \ block_statements, \ input_rule, \ argument_for_input, \ output_rule, \ \ program_rule, \ \ non_zero_digit, \ digit__iteration, \ digit, \ unsigned_value, \ value, \ \ sign, \ ident, \ letter_in_upper_case, \ letter_in_lower_case, \ sign_plus, \ sign_minus	#define NONTERMINALS labeled_point, \ goto_label, \ program_name, \ value_type, \ array_specify, \ declaration_element, \ array_specify__optional, \ other_declaration_ident, \ declaration, \ other_declaration_ident__iteration, \ index_action, \ unary_operator, \ unary_operation, \ binary_operator, \ binary_action, \ left_expression, \ group_expression, \ index_action__optional, \ expression, \ binary_action__iteration, \ expression_or_cond_block__with_optional_assign, \ assign_to_right, \ assign_to_right__optional, \ if_expression, \ body_for_true, \ false_cond_block_without_else, \ body_for_false, \ cond_block, \ false_cond_block_without_else__iteration, \ body_for_false__optional, \ statement, \ block_statements, \ input_rule, \ argument_for_input, \ output_rule, \ statement__iteration, \ expression__optional, \ program_rule, \ declaration__optional, \ non_zero_digit, \ digit__iteration, \ digit, \ unsigned_value, \ value, \ sign__optional, \ sign, \ ident, \ letter_in_upper_case, \ letter_in_lower_case, \ sign_plus, \ sign_minus	
		T = { "." "GOTO", "INTEGER16", " " "NOT", "AND", "OR", "==" "!=", "<=", ">=", "+", "-", "*", "/", "DIV", "MOD", "(", ")", "::", "ELSE", "IF", "EXIT", "GET", "PUT", "NAME", "BODY", "DATA", "BEGIN",	T = { "." "GOTO", "INTEGER16", " " "NOT", "AND", "OR", "==" "!=", "<=", ">=", "+", "-", "*", "/", "DIV", "MOD", "(", ")", "::", "ELSE", "IF", "EXIT", "GET", "PUT", "NAME", "BODY", "DATA", "BEGIN",	#define TOKENS \ tokenCOLON, \ tokenGOTO, \ tokenINTEGER16, \ tokenCOMMA, \ tokenNOT, \ tokenAND, \ tokenOR, \ tokenEQUAL, \ tokenNOTEQUAL, \ tokenLESSOREQUAL, \ tokenGREATEROREQUAL, \ tokenPLUS, \ tokenMINUS, \ tokenMUL, \ tokenDIV, \ tokenMOD, \ tokenGROUPEXPRESSIONBEGIN, \ tokenGROUPEXPRESSIONEND, \ tokenLRASSIGN, \ tokenELSE, \ tokenIF, \ tokenEXIT, \ tokenGET, \ tokenPUT, \ tokenNAME, \ tokenBODY, \ tokenDATA, \ tokenBEGIN, \ 	#define TOKENS \ tokenCOLON, \ tokenGOTO, \ tokenINTEGER16, \ tokenCOMMA, \ tokenNOT, \ tokenAND, \ tokenOR, \ tokenEQUAL, \ tokenNOTEQUAL, \ tokenLESSOREQUAL, \ tokenGREATEROREQUAL, \ tokenPLUS, \ tokenMINUS, \ tokenMUL, \ tokenDIV, \ tokenMOD, \ tokenGROUPEXPRESSIONBEGIN, \ tokenGROUPEXPRESSIONEND, \ tokenLRASSIGN, \ tokenELSE, \ tokenIF, \ tokenEXIT, \ tokenGET, \ tokenPUT, \ tokenNAME, \ tokenBODY, \ tokenDATA, \ tokenBEGIN, \ 	

						#define T_COLON_3 ""
				tokenGOTO = "GOTO" >> STRICT_BOUNDARIES;	tokenGOTO = "GOTO" >> STRICT_BOUNDARIES;	#define T_GOTO_0 "GOTO" #define T_GOTO_1 "" #define T_GOTO_2 "" #define T_GOTO_3 ""
				tokenINTEGER16 = "INTEGER16" >> STRICT_BOUNDARIES;	tokenINTEGER16 = "INTEGER16" >> STRICT_BOUNDARIES;	#define T_DATA_TYPE_0 "INTEGER16" #define T_DATA_TYPE_1 "" #define T_DATA_TYPE_2 "" #define T_DATA_TYPE_3 ""
				tokenCOMMA = ",", >> BOUNDARIES;	tokenCOMMA = ",", >> BOUNDARIES;	#define T_COMA_0 "," #define T_COMA_1 "" #define T_COMA_2 "" #define T_COMA_3 ""
						#define T_BITWISE_NOT_0 "~" #define T_BITWISE_NOT_1 "" #define T_BITWISE_NOT_2 "" #define T_BITWISE_NOT_3 ""
				tokenNOT = "NOT" >> STRICT_BOUNDARIES;	tokenNOT = "NOT" >> STRICT_BOUNDARIES;	#define T_NOT_0 "NOT" #define T_NOT_1 "" #define T_NOT_2 "" #define T_NOT_3 ""
						#define T_BITWISE_AND_0 "&" #define T_BITWISE_AND_1 "" #define T_BITWISE_AND_2 "" #define T_BITWISE_AND_3 ""
				tokenAND = "AND" >> STRICT_BOUNDARIES;	tokenAND = "AND" >> STRICT_BOUNDARIES;	#define T_AND_0 "AND" #define T_AND_1 "" #define T_AND_2 "" #define T_AND_3 ""
						#define T_BITWISE_OR_0 " " #define T_BITWISE_OR_1 "" #define T_BITWISE_OR_2 "" #define T_BITWISE_OR_3 ""
				tokenOR = "OR" >> STRICT_BOUNDARIES;	tokenOR = "OR" >> STRICT_BOUNDARIES;	#define T_OR_0 "OR" #define T_OR_1 "" #define T_OR_2 "" #define T_OR_3 ""
				tokenEQUAL = "==" >> BOUNDARIES;	tokenEQUAL = "==" >> BOUNDARIES;	#define T_EQUAL_0 "==" #define T_EQUAL_1 "" #define T_EQUAL_2 "" #define T_EQUAL_3 ""
				tokenNOTEQUAL = "!=" >> BOUNDARIES;	tokenNOTEQUAL = "!=" >> BOUNDARIES;	#define T_NOT_EQUAL_0 "!=" #define T_NOT_EQUAL_1 "" #define T_NOT_EQUAL_2 "" #define T_NOT_EQUAL_3 ""
				tokenLESSOREQUAL = "<=" >> BOUNDARIES;	tokenLESSOREQUAL = "<=" >> BOUNDARIES;	#define T_LESS_OR_EQUAL_0 "<="
				tokenGREATEROREQUAL = ">=" >> BOUNDARIES;	tokenGREATEROREQUAL = ">=" >> BOUNDARIES;	#define T_GREATER_OR_EQUAL_0 ">="
				tokenPLUS = "+" >> BOUNDARIES;	tokenPLUS = "+" >> BOUNDARIES;	#define T_ADD_0 "+" #define T_ADD_1 "" #define T_ADD_2 "" #define T_ADD_3 ""
				tokenMINUS = "-", >> BOUNDARIES;	tokenMINUS = "-", >> BOUNDARIES;	#define T_SUB_0 "-" #define T_SUB_1 "" #define T_SUB_2 "" #define T_SUB_3 ""
				tokenMUL = "*" >> BOUNDARIES;	tokenMUL = "*" >> BOUNDARIES;	#define T_MUL_0 "*"
				tokenDIV = "DIV" >> STRICT_BOUNDARIES;	tokenDIV = "DIV" >> STRICT_BOUNDARIES;	#define T_DIV_0 "DIV" #define T_DIV_1 "" #define T_DIV_2 "" #define T_DIV_3 ""
				tokenMOD = "MOD" >> STRICT_BOUNDARIES;	tokenMOD = "MOD" >> STRICT_BOUNDARIES;	#define T_MOD_0 "MOD" #define T_MOD_1 "" #define T_MOD_2 "" #define T_MOD_3 ""
				tokenLRASSIGN = "=: " >> BOUNDARIES;	tokenLRASSIGN = "=: " >> BOUNDARIES;	#define T_LRASSIGN_0 "=: " #define T_LRASSIGN_1 "" #define T_LRASSIGN_2 "" #define T_LRASSIGN_3 ""
						#define T_THEN_BLOCK_0 "{" #define T_THEN_BLOCK_1 "" #define T_THEN_BLOCK_2 "" #define T_THEN_BLOCK_3 ""
				tokenELSE = "ELSE" >> STRICT_BOUNDARIES;	tokenELSE = "ELSE" >> STRICT_BOUNDARIES;	#define T_ELSE_BLOCK_0 "ELSE" #define T_ELSE_BLOCK_1 T_BEGIN_BLOCK_0 #define T_ELSE_BLOCK_2 "" #define T_ELSE_BLOCK_3 ""
				tokenIF = "IF" >> STRICT_BOUNDARIES;	tokenIF = "IF" >> STRICT_BOUNDARIES;	#define T_IF_0 "IF" #define T_IF_1 "" #define T_IF_2 "" #define T_IF_3 ""
						#define T_ELSE_IF_0 "ELSE" #define T_ELSE_IF_1 T_IF_0 #define T_ELSE_IF_2 "" #define T_ELSE_IF_3 ""
				tokenEXIT = "EXIT" >> STRICT_BOUNDARIES;	tokenEXIT = "EXIT" >> STRICT_BOUNDARIES;	#define T_EXIT_0 "EXIT" #define T_EXIT_1 "" #define T_EXIT_2 "" #define T_EXIT_3 ""
				tokenGET = "GET" >> STRICT_BOUNDARIES;	tokenGET = "GET" >> STRICT_BOUNDARIES;	#define T_INPUT_0 "GET" #define T_INPUT_1 "" #define T_INPUT_2 "" #define T_INPUT_3 ""
				tokenPUT = "PUT" >> STRICT_BOUNDARIES;	tokenPUT = "PUT" >> STRICT_BOUNDARIES;	#define T_OUTPUT_0 "PUT" #define T_OUTPUT_1 "" #define T_OUTPUT_2 "" #define T_OUTPUT_3 ""
				tokenNAME = "NAME" >> STRICT_BOUNDARIES;	tokenNAME = "NAME" >> STRICT_BOUNDARIES;	#define T_NAME_0 "NAME"

						#define T_NAME_1 "" #define T_NAME_2 "" #define T_NAME_3 ""
				tokenBODY = "BODY" >> STRICT_BOUNDARIES;	tokenBODY = "BODY" >> STRICT_BOUNDARIES;	#define T_BODY_0 "BODY" #define T_BODY_1 "" #define T_BODY_2 "" #define T_BODY_3 ""
				tokenDATA = "DATA" >> STRICT_BOUNDARIES;	tokenDATA = "DATA" >> STRICT_BOUNDARIES;	#define T_DATA_0 "DATA" #define T_DATA_1 "" #define T_DATA_2 "" #define T_DATA_3 ""
				tokenBEGIN = "BEGIN" >> STRICT_BOUNDARIES;	tokenBEGIN = "BEGIN" >> STRICT_BOUNDARIES;	#define T_BEGIN_0 "BEGIN" #define T_BEGIN_1 "" #define T_BEGIN_2 "" #define T_BEGIN_3 ""
				tokenEND = "END" >> STRICT_BOUNDARIES;	tokenEND = "END" >> STRICT_BOUNDARIES;	#define T_END_0 "END" #define T_END_1 "" #define T_END_2 "" #define T_END_3 ""
						#define T_NULL_STATEMENT_0 "NULL" #define T_NULL_STATEMENT_1 "STATEMENT" #define T_NULL_STATEMENT_2 "" #define T_NULL_STATEMENT_3 ""
						#define GRAMMAR_LL2__2025 {\
labeled_point = ident , ":";	labeled_point = ident , ":";	labeled_point → ident ":"	labeled_point(1: "ident_terminal") → ident ":"	labeled_point = ident >> tokenCOLON;	labeled_point = ident >> tokenCOLON;	{ LA_IS, ("ident_terminal"), { "labeled_point", {\
goto_label = "GOTO" , ident;	goto_label = "GOTO" , ident;	goto_label → "GOTO" ident	goto_label(1: "GOTO") → "GOTO" ident	goto_label = tokenGOTO >> ident;	goto_label = tokenGOTO >> ident;	{ LA_IS, (T_GOTO_0), { "goto_label", {\
program_name = ident;	program_name = ident;	program_name → ident	program_name(1: "ident_terminal") → ident	program_name = SAME_RULE(ident);	program_name = SAME_RULE(ident);	{ LA_IS, ("ident_terminal"), { "program_name", {\
value_type = "INTEGER16";	value_type = "INTEGER16";	value_type → "INTEGER16"	value_type(1: "INTEGER16") → "INTEGER16"	value_type = SAME_RULE(tokenINTEGER16);	value_type = SAME_RULE(tokenINTEGER16);	{ LA_IS, (T_DATA_TYPE_0), { "value_type", {\
	array_specify = "[", unsigned_value, "]"	array_specify → "[" unsigned_value "]"	array_specify(1: "["") → "[" unsigned_value "]"		array_specify = "[" >> unsigned_value >> "]"	{ LA_IS, ("["), { "array_specify", {\
declaration_element = ident, array_specify__optional;	declaration_element = ident, array_specify__optional;	declaration_element → ident array_specify__optional	declaration_element(1: "ident_terminal") → ident array_specify__optional	declaration_element = ident >> -(tokenLEFTSQUAREBRACKETS >> unsigned_value >> tokenRIGHTSQUAREBRACKETS);	declaration_element = ident >> array_specify__optional;	{ LA_IS, ("ident_terminal"), { "declaration_element", {\
	array_specify__optional = array_specify ε;	array_specify__optional → array_specify array_specify__optional → ε	array_specify__optional(1: "["") → array_specify array_specify__optional(1: !"[") → ε		array_specify__optional = array_specify "";	{ LA_IS, ("["), { "array_specify__optional", {\
other_declaration_ident = " ,", declaration_element;	other_declaration_ident = " ,", declaration_element;	other_declaration_ident → " ,", declaration_element	other_declaration_ident(1: " ,") → " ,", declaration_element	other_declaration_ident = tokenCOMMA >> declaration_element;	other_declaration_ident = tokenCOMMA >> declaration_element;	{ LA_IS, (T_COMA_0), { "other_declaration_ident", {\
declaration = value_type, declaration_element , other_declaration_ident__iteration;	declaration = value_type, declaration_element , other_declaration_ident__iteration;	declaration → value_type declaration_element other_declaration_ident__iteration	declaration(1: "INTEGER16") → value_type declaration_element other_declaration_ident__iteration	declaration = value_type >> declaration_element >> *other_declaration_ident;	declaration = value_type >> declaration_element >> other_declaration_ident__iteration;	{ LA_IS, (T_DATA_TYPE_0), { "declaration", {\
	other_declaration_ident__iteration = other_declaration_ident, other_declaration_ident__iteration ε;	other_declaration_ident__iteration → other_declaration_ident other_declaration_ident__iteration false_cond_block_without_else__iteration → ε	other_declaration_ident__iteration(1: " ,") → other_declaration_ident other_declaration_ident__iteration false_cond_block_without_else__iteration(1: !",") → ε		other_declaration_ident__iteration = other_declaration_ident >> other_declaration_ident__iteration "";	{ LA_IS, (T_COMA_0), {
index_action = "[", expression , "]"	index_action = "[", expression , "]"	index_action → "[" expression "]"	index_action(1: "["") → "[" expression "]"	index_action = tokenLEFTSQUAREBRACKETS >> expression >> tokenRIGHTSQUAREBRACKETS;	index_action = tokenLEFTSQUAREBRACKETS >> expression >> tokenRIGHTSQUAREBRACKETS;	{ LA_IS, ("["), { "index_action", {\
unary_operator = "NOT";	unary_operator = "NOT";	unary_operator → "NOT"	unary_operator(1: "NOT") → "NOT"	unary_operator = SAME_RULE(tokenNOT);	unary_operator = SAME_RULE(tokenNOT);	{ LA_IS, (T_NOT_0), { "unary_operator", {\
unary_operation = unary_operator , expression;	unary_operation = unary_operator , expression;	unary_operation → unary_operator expression	unary_operation(1: "NOT") → unary_operator expression	unary_operation = unary_operator >> expression;	unary_operation = unary_operator >> expression;	{ LA_IS, (T_NOT_0), { "unary_operation", {\
binary_operator = "AND" "OR" "=" "!=" "<=" ">=" "+" "-" "*" "DIV" "MOD";	binary_operator = "AND" "OR" "=" "!=" "<=" ">=" "+" "-" "*" "DIV" "MOD";	binary_operator → "AND" binary_operator → "OR" binary_operator → "=" binary_operator → "!=" binary_operator → "<=" binary_operator → ">=" binary_operator → "+" binary_operator → "-" binary_operator → "*" binary_operator → "DIV" binary_operator → "MOD"	binary_operator(1: "AND") → "AND" binary_operator(1: "OR") → "OR" binary_operator(1: "=") → "=" binary_operator(1: "!=") → "!=" binary_operator(1: "<=") → "<=" binary_operator(1: ">=") → ">=" binary_operator(1: "+") → "+" binary_operator(1: "-") → "-" binary_operator(1: "*") → "*" binary_operator(1: "DIV") → "DIV" binary_operator(1: "MOD") → "MOD"	binary_operator = tokenAND tokenOR tokenEQUAL tokenNOTEQUAL tokenLESSOREQUAL tokenGREATEROREQUAL tokenPLUS tokenMINUS tokenMUL tokenDIV tokenMOD;	binary_operator = tokenAND tokenOR tokenEQUAL tokenNOTEQUAL tokenLESSOREQUAL tokenGREATEROREQUAL tokenPLUS tokenMINUS tokenMUL tokenDIV tokenMOD;	{ LA_IS, (T_AND_0), { "binary_operator", {\
binary_operator = "AND" "OR" "=" "!=" "<=" ">=" "+" "-" "*" "DIV" "MOD";	binary_operator = "AND" "OR" "=" "!=" "<=" ">=" "+" "-" "*" "DIV" "MOD";	binary_operator → "AND" binary_operator → "OR" binary_operator → "=" binary_operator → "!=" binary_operator → "<=" binary_operator → ">=" binary_operator → "+" binary_operator → "-" binary_operator → "*" binary_operator → "DIV" binary_operator → "MOD"	binary_operator(1: "AND") → "AND" binary_operator(1: "OR") → "OR" binary_operator(1: "=") → "=" binary_operator(1: "!=") → "!=" binary_operator(1: "<=") → "<=" binary_operator(1: ">=") → ">=" binary_operator(1: "+") → "+" binary_operator(1: "-") → "-" binary_operator(1: "*") → "*" binary_operator(1: "DIV") → "DIV" binary_operator(1: "MOD") → "MOD"	binary_operator = tokenAND tokenOR tokenEQUAL tokenNOTEQUAL tokenLESSOREQUAL tokenGREATEROREQUAL tokenPLUS tokenMINUS tokenMUL tokenDIV tokenMOD;	binary_operator = tokenAND tokenOR tokenEQUAL tokenNOTEQUAL tokenLESSOREQUAL tokenGREATEROREQUAL tokenPLUS tokenMINUS tokenMUL tokenDIV tokenMOD;	{ LA_IS, (T_AND_0), { "binary_operator", {\

	binary_action = binary_operator , expression;	binary_action → binary_operator expression	binary_action(1: "AND", "OR", "=", "!=", "<=", ">=", "+", "-", "**", "MOD") → binary_operator expression		binary_action = binary_operator >> expression;	{ LA_IS, { T_AND_0, T_OR_0, T_EQUAL_0, T_NOT_EQUAL_0, T_LESS_OR_EQUAL_0, T_GREATER_OR_EQUAL_0, T_ADD_0, T_SUB_0, T_MUL_0, T_DIV_0, T_MOD_0 }, { "binary_action", {\ LA_IS, { "" }, 2, { "binary_operator", "expression" } } } } \\\n
binary_action = binary_operator , expression;				binary_action = binary_operator >> expression;		
left_expression = group_expression unary_operation ident , [index_action] value cond_block;	left_expression = group_expression unary_operation ident , index_action__optional value cond_block;	left_expression → group_expression left_expression → unary_operation left_expression → ident , index_action__optional left_expression → value left_expression → cond_block	left_expression(1: "(" → group_expression left_expression(1: "NOT") → unary_operation left_expression(1: "_" → ident , index_action__optional left_expression(1: "0", "1", "2", "3", "4", "5", "6", "7", "8", "9") → value left_expression(1: "+", "-", ":", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9") → value left_expression(1: "IF") → cond_block	left_expression = group_expression unary_operation ident >> -index_action value cond_block;	left_expression = group_expression unary_operation ident >> index_action__optional value cond_block;	{ LA_IS, { "(" }, { "left_expression", {\ LA_IS, { "" }, 1, { "group_expression" } } } } \\\n{ LA_IS, { T_NOT_0 }, { "left_expression", {\ LA_IS, { "" }, 1, { "unary_operation" } } } } \\\n{ LA_IS, { "ident_terminal" }, { "left_expression", {\ LA_IS, { "" }, 2, { "ident", "index_action__optional" } } } } \\\n{ LA_IS, { "unsigned_value_terminal" }, { "left_expression", {\ LA_IS, { "" }, 1, { "value" } } } } \\\n{ LA_IS, { T_ADD_0, T_SUB_0 }, { "left_expression", {\ LA_IS, { "unsigned_value_terminal" }, 1, { "value" } } } } \\\n/*{ LA_NOT, { "unsigned_value_terminal" }, 1, { "unary_operation" } } */ \\\n{ LA_IS, { T_IF_0 }, { "left_expression", {\ LA_IS, { "" }, 1, { "cond_block" } } } } \\\n
	index_action__optional = index_action ε;	index_action__optional → index_action index_action__optional → ε	index_action__optional(1: "(" → index_action index_action__optional(1: "!" → ε		index_action__optional = index_action "";	{ LA_IS, { "(" }, { "index_action__optional", {\ LA_IS, { "" }, 1, { "index_action" } } } } \\\n{ LA_NOT, { "(" }, { "index_action__optional", {\ LA_IS, { "" }, 0, { "" } } } } \\\n
expression = left_expression , {binary_action};	expression = left_expression , binary_action__iteration;	expression → left_expression binary_action__iteration	expression(1: "(" , "NOT", "+", "-", ":", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → left_expression binary_action__iteration	expression = left_expression >> *binary_action;	expression = left_expression >> binary_action__iteration;	{ LA_IS, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "expression", {\ LA_IS, { "" }, 2, { "left_expression", "binary_action__iteration" } } } } \\\n
	binary_action__iteration = binary_action, binary_action__iteration ε;	binary_action__iteration → binary_action binary_action__iteration → ε	binary_action__iteration(1: "AND", "OR", "=", "!=", "<=", ">=", "+", "-", "**", "DIV", "MOD") → binary_action binary_action__iteration binary_action__iteration(1: !"AND", !"OR", !"=", !"!=, !"<=", !">=", !"+, !"-, !":, !"0", !"1", !"2", !"3", !"4", !"5", !"6", !"7", !"8", !"9", !"IF") → ε		binary_action__iteration = binary_action >> binary_action__iteration "";	{ LA_IS, { T_AND_0, T_OR_0, T_EQUAL_0, T_NOT_EQUAL_0, T_LESS_OR_EQUAL_0, T_GREATER_OR_EQUAL_0, T_ADD_0, T_SUB_0, T_MUL_0, T_DIV_0, T_MOD_0 }, { "binary_action__iteration", {\ LA_IS, { "" }, 2, { "binary_action", "binary_action__iteration" } } } } \\\n{ LA_NOT, { T_AND_0, T_OR_0, T_EQUAL_0, T_NOT_EQUAL_0, T_LESS_OR_EQUAL_0, T_GREATER_OR_EQUAL_0, T_ADD_0, T_SUB_0, T_MUL_0, T_DIV_0, T_MOD_0 }, { "binary_action__iteration", {\ LA_IS, { "" }, 0, { "" } } } } \\\n
group_expression = "(" , expression , ")";	group_expression = "(" , expression , ")";	group_expression → "(" expression ")"	group_expression(1: "(" → "(" expression ")"	group_expression = tokenGROUPEXPRESSSIONBEGIN >> expression >> tokenGROUPEXPRESSSIONEND;	group_expression = tokenGROUPEXPRESSSIONBEGIN >> expression >> tokenGROUPEXPRESSSIONEND;	{ LA_IS, { "(" }, { "group_expression", {\ LA_IS, { "" }, 3, { "(" , "expression", ")" } } } } \\\n
expression_or_cond_block__with_optional_assign = expression , assign_to_right__optional;	expression_or_cond_block__with_optional_assign = expression , assign_to_right__optional;	expression_or_cond_block__with_optional_assign → expression assign_to_right__optional	expression_or_cond_block__with_optional_assign(1: "(" , "NOT", "+", "-", ":", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression assign_to_right__optional	expression_or_cond_block__with_optional_assign = expression >> -(tokenLRASSIGN >> ident >> -index_action);	expression_or_cond_block__with_optional_assign = expression >> assign_to_right__optional;	{ LA_IS, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "expression_or_cond_block__with_optional_assign", {\ LA_IS, { "" }, 2, { "expression", "assign_to_right__optional" } } } } \\\n
	assign_to_right = "=", ident , index_action__optional;	assign_to_right → "=" ident index_action__optional	assign_to_right(1: "=" → "=" ident index_action__optional		assign_to_right = tokenLRASSIGN >> ident >> index_action__optional;	{ LA_IS, { T_LRASSIGN_0 }, { "assign_to_right", {\ LA_IS, { "" }, 3, { T_LRASSIGN_0, "ident", "index_action__optional" } } } } \\\n
	assign_to_right__optional = assign_to_right ε;	assign_to_right__optional → assign_to_right assign_to_right__optional → ε;	assign_to_right__optional(1: "=" → assign_to_right assign_to_right__optional(1: !"=" → ε;		assign_to_right__optional = assign_to_right "";	{ LA_IS, { T_LRASSIGN_0 }, { "assign_to_right__optional", {\ LA_IS, { "" }, 1, { "assign_to_right" } } } } \\\n{ LA_NOT, { T_LRASSIGN_0 }, { "assign_to_right__optional", {\ LA_IS, { "" }, 0, { "" } } } } \\\n
if_expression = expression;	if_expression = expression;	if_expression → expression	if_expression(1: "(" , "NOT", "+", "-", ":", "0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "IF") → expression	if_expression = SAME_RULE(expression);	if_expression = SAME_RULE(expression);	{ LA_IS, { "(" , T_NOT_0, T_ADD_0, T_SUB_0, "ident_terminal", "unsigned_value_terminal", T_IF_0 }, { "if_expression", {\ LA_IS, { "" }, 1, { "expression" } } } } \\\n
body_for_true = block_statements;	body_for_true = block_statements;	body_for_true → block_statements	body_for_true(1: "{" → block_statements	body_for_true = SAME_RULE(block_statements);	body_for_true = SAME_RULE(block_statements);	{ LA_IS, { T_BEGIN_BLOCK_0 }, { "body_for_true", {\ LA_IS, { "" }, 1, { "block_statements" } } } } \\\n
false_cond_block_without_else = "ELSE" , "IF" , if_expression , body_for_true;	false_cond_block_without_else = "ELSE" , "IF" , if_expression , body_for_true;	false_cond_block_without_else → "ELSE" "IF" if_expression body_for_true	false_cond_block_without_else(1: "ELSE") → "ELSE" "IF" if_expression body_for_true	false_cond_block_without_else = tokenELSE >> tokenIF >> if_expression >> body_for_true;	false_cond_block_without_else = tokenELSE >> tokenIF >> if_expression >> body_for_true;	{ LA_IS, { T_ELSE_IF_0 }, { "false_cond_block_without_else", {\ LA_IS, { "" }, 4, { T_ELSE_IF_0, T_ELSE_IF_1, "if_expression", "body_for_true" } } } } \\\n
body_for_false = "ELSE" , block_statements;	body_for_false = "ELSE" , block_statements;	body_for_false → "ELSE" block_statements	body_for_false(1: "ELSE") → "ELSE" block_statements	body_for_false = tokenELSE >> block_statements;	body_for_false = tokenELSE >> block_statements;	{ LA_IS, { T_ELSE_BLOCK_0 }, { "body_for_false", {\ LA_IS, { "" }, 2, { T_ELSE_BLOCK_0, "block_statements" } } } } \\\n
cond_block = "IF" , if_expression , body_for_true, { false_cond_block_without_else } , {body_for_false};	cond_block = "IF" , if_expression , body_for_true, { false_cond_block_without_else } , {body_for_false};	cond_block → "IF" if_expression body_for_true false_cond_block_without_else__iteration body_for_false__optional	cond_block(1: "IF") → "IF" if_expression body_for_true false_cond_block_without_else__iteration body_for_false__optional	cond_block = tokenIF >> if_expression >> body_for_true >> *false_cond_block_without_else >> -body_for_false;	cond_block = tokenIF >> if_expression >> body_for_true >> false_cond_block_without_else__iteration >> body_for_false__optional;	{ LA_IS, { T_IF_0 }, { "cond_block", {\ LA_IS, { "" }, 5, { T_IF_0, "if_expression", "body_for_true", "false_cond_block_without_else__iteration", "body_for_false__optional" } } } } \\\n
	false_cond_block_without_else__iteration = false_cond_block_without_else, false_cond_block_without_else__iteration ε;	false_cond_block_without_else__iteration → false_cond_block_without_else false_cond_block_without_else__iteration → ε	false_cond_block_without_else__iteration(1: "ELSE"; 2: !"IF") → false_cond_block_without_else false_cond_block_without_else__iteration(1: "ELSE"; 2: !"IF") → ε false_cond_block_without_else__iteration(1: !"ELSE") → ε		false_cond_block_without_else__iteration = false_cond_block_without_else >> false_cond_block_without_else__iteration "";	{ LA_IS, { T_ELSE_IF_0 }, { "false_cond_block_without_else__iteration", {\ LA_IS, { T_ELSE_IF_1 }, 2, { "false_cond_block_without_else", "false_cond_block_without_else__iteration" } } } } \\\n{ LA_NOT, { T_ELSE_IF_0 }, { "false_cond_block_without_else__iteration", {\ LA_IS, { "" }, 0, { "" } } } } \\\n

						}}}\
	body_for_false__optional = body_for_false ε;	body_for_false__optional → body_for_false body_for_false__optional → ε	body_for_false__optional(1: "FALSE") → body_for_false body_for_false__optional(1: !"FALSE") → ε		body_for_false__optional = body_for_false "";	{LA_IS, {T_ELSE_BLOCK_0}, { "body_for_false__optional", {\ {LA_IS, {""}, 1, { "body_for_false" }}\ }}}\
input_rule = "GET" , (ident , [index_action] "(" , ident , [index_action] , ")");	input_rule = "GET", argument_for_input;	input_rule → "GET" argument_for_input	input_rule(1: "GET") → "GET" argument_for_input	input_rule = tokenGET >> (ident >> -index_action tokenGROUPEXPRESSIONBEGIN >> ident >> -index_action >> tokenGROUPEXPRESSIONEND);	input_rule = tokenGET >> argument_for_input;	{LA_IS, {T_INPUT_0}, { "input_rule", {\ {LA_IS, {""}, 2, {T_INPUT_0, "argument_for_input" }}\ }}}\
	argument_for_input = ident , index_action__optional; argument_for_input = "(" , "ident" , "index_action__optional" , ")" ;	argument_for_input → ident index_action__optional argument_for_input → "(" "ident" "index_action__optional" ")"	argument_for_input(1: " _ ") → ident index_action__optional argument_for_input(1: "(") → "(" "ident" "index_action__optional" ")"		argument_for_input = ident >> index_action__optional tokenGROUPEXPRESSIONBEGIN >> ident >> index_action__optional >> tokenGROUPEXPRESSIONEND;	{LA_IS, { "ident_terminal" }, { "argument_for_input", {\ {LA_IS, {""}, 2, { "ident", "index_action__optional" }}\ }}}\
						{LA_IS, { "" }, { "argument_for_input", {\ {LA_IS, {""}, 4, { "(" , "ident", "index_action__optional" , ")" }}\ }}}\
output_rule = "PUT" , expression;	output_rule = "PUT", expression;	output_rule → "PUT" expression	output(1: "PUT") → "PUT" expression	output_rule = tokenPUT >> expression;	output_rule = tokenPUT >> expression;	{LA_IS, {T_OUTPUT_0}, { "output_rule", {\ {LA_IS, {""}, 2, {T_OUTPUT_0, "expression" }}\ }}}\
statement = labeled_point expression_or_cond_block__with_optional_assign goto_label input_rule output_rule ";;";	statement = labeled_point expression_or_cond_block__with_optional_assign goto_label input_rule output_rule ";;";	statement → labeled_point statement → expression_or_cond_block__with_optional_assign statement → goto_label statement → input_rule statement → output_rule statement → ";;"	statement(1: " _ " , " _ .") → labeled_point statement(1: " _ " ; 2: !" _ ") → expression_or_cond_block__with_optional_assign statement(1: "(" , "NOT" , "+" , " _ " , "0" , "1" , "2" , "3" , "4" , "5" , "6" , "7" , "8" , "9" , "IF") → expression_or_cond_block__with_optional_assign statement(1: "GOTO") → goto_label statement(1: "GET") → input_rule statement(1: "PUT") → output_rule statement(1: " _ .") → " _ ;"	statement = labeled_point /* labeled_point first* / expression_or_cond_block__with_optional_assign goto_label input_rule output_rule tokenSEMICOLON;	statement = labeled_point /* labeled_point first* / expression_or_cond_block__with_optional_assign goto_label input_rule output_rule tokenSEMICOLON;	{LA_IS, { "ident_terminal" }, { "statement", {\ {LA_IS, {T_COLON_0}, 1, { "labeled_point" }}\, {LA_NOT, {T_COLON_0}, 1, { "expression_or_cond_block__with_optional_assign" }}\ }}}\
						{LA_IS, { "(" , T_NOT_0 , "unsigned_value_terminal", T_ADD_0 , T_SUB_0 , T_IF_0 }, { "statement", {\ {LA_IS, {""}, 1, { "expression_or_cond_block__with_optional_assign" }}\ }}}\
						{LA_IS, {T_GOTO_0}, { "statement", {\ {LA_IS, {""}, 1, { "goto_label" }}\ }}}\
						{LA_IS, {T_INPUT_0}, { "statement", {\ {LA_IS, {""}, 1, { "input_rule" }}\ }}}\
						{LA_IS, {T_OUTPUT_0}, { "statement", {\ {LA_IS, {""}, 1, { "output_rule" }}\ }}}\
						{LA_IS, {T_SEMICOLON_0}, { "statement", {\ {LA_IS, {""}, 1, { ";" }}\ }}}\
	statement__iteration = statement, statement__iteration ε;	statement__iteration → statement statement__iteration statement__iteration → ε	statement__iteration(1: " _ " , "(" , "NOT" , "0" , "1" , "2" , "3" , "4" , "5" , "6" , "7" , "8" , "9" , "+" , " _ " , "IF" , "GOTO" , "GET" , "PUT" , " _ .") → statement statement__iteration statement__iteration(1: !" _ " , ! "(" , ! "NOT" , ! "0" , ! "1" , ! "2" , ! "3" , ! "4" , ! "5" , ! "6" , ! "7" , ! "8" , ! "9" , ! "+" , ! " _ " , ! "IF" , ! "GOTO" , ! "GET" , ! "PUT" , ! " _ .") → ε		statement__iteration = statement >> statement__iteration "";	{LA_IS, { "ident_terminal" , "(" , T_NOT_0 , "unsigned_value_terminal" , T_ADD_0 , T_SUB_0 , T_IF_0 , T_GOTO_0 , T_INPUT_0 , T_OUTPUT_0 , T_SEMICOLON_0 }, { "statement__iteration", {\ {LA_IS, {""}, 2, { "statement", "statement__iteration" }}\ }}}\
						{LA_NOT, { "ident_terminal" , "(" , T_NOT_0 , "unsigned_value_terminal" , T_ADD_0 , T_SUB_0 , T_IF_0 , T_GOTO_0 , T_INPUT_0 , T_OUTPUT_0 , T_SEMICOLON_0 }, { "statement__iteration", {\ {LA_IS, {""}, 0, { "" }}\ }}}\
block_statements = "(" , {statement} , ")" ;	block_statements = "(" , statement__iteration , ")" ;	block_statements → "(" statement__iteration ")"	block_statements(1: "(") → "(" statement__iteration ")"	block_statements = tokenBEGINBLOCK >> *statement >> tokenENDBLOCK;	block_statements = tokenBEGINBLOCK >> statement__iteration >> tokenENDBLOCK;	{LA_IS, {T_BEGIN_BLOCK_0}, { "block_statements", {\ {LA_IS, {""}, 3, {T_BEGIN_BLOCK_0, "statement__iteration" , T_END_BLOCK_0 }}\ }}}\
	expression__optional = expression "";	expression__optional → expression expression__optional → ε	expression__optional(1: "(" , "NOT" , "+" , " _ " , "0" , "1" , "2" , "3" , "4" , "5" , "6" , "7" , "8" , "9" , "IF") → expression expression__optional(1: ! "(" , ! "NOT" , ! "+" , ! " _ " , ! "0" , ! "1" , ! "2" , ! "3" , ! "4" , ! "5" , ! "6" , ! "7" , ! "8" , ! "9" , ! "IF") → ε		expression__optional = expression "";	{LA_IS, { "(" , T_NOT_0 , T_ADD_0 , T_SUB_0 , "ident_terminal" , "unsigned_value_terminal" , T_IF_0 , { "expression__optional", {\ {LA_IS, {""}, 1, { "expression" }}\ }}\
						{LA_NOT, { "(" , T_NOT_0 , T_ADD_0 , T_SUB_0 , "ident_terminal" , "unsigned_value_terminal" , T_IF_0 , { "expression__optional", {\ {LA_IS, {""}, 0, { "" }}\ }}\
program_rule = "NAME" , program_name , " , " , "BODY" , "DATA" , [declaration] , " , " , {statement} , "END" ;	program_rule = "NAME" , program_name , " , " , "BODY" , "DATA" , declaration__optional , " , " , statement__iteration , "END" ;	program_rule → "NAME" program_name " , " , "BODY" "DATA" declaration__optional " , " , statement__iteration "END"	program_rule(1: "NAME") → "NAME" program_name " , " , "BODY" "DATA" declaration__optional " , " , statement__iteration "END"	program_rule = BOUNDARIES >> tokenNAME >> program_name >> tokenSEMICOLON >> tokenBODY >> tokenDATA >> (- declaration) >> tokenSEMICOLON >> *statement >> tokenEND;	program_rule = BOUNDARIES >> tokenNAME >> program_name >> tokenSEMICOLON >> tokenBODY >> tokenDATA >> declaration__optional >> tokenSEMICOLON >> statement__iteration >> tokenEND;	{LA_IS, {T_NAME_0}, { "program_rule", {\ {LA_IS, {""}, 9, {T_NAME_0, "program_name", T_SEMICOLON_0 , T_BODY_0 , T_DATA_0 , "declaration__optional" , T_SEMICOLON_0 , "statement__iteration" , T_END_0 }}\ }}}\
	declaration__optional = declaration "";	declaration__optional → declaration declaration__optional → ε	declaration__optional(1: "INTEGER16") → declaration declaration__optional(1: !"INTEGER16") → ε		declaration__optional = declaration "";	{LA_IS, {T_DATA_TYPE_0}, { "declaration__optional", {\ {LA_IS, {""}, 1, { "declaration" }}\ }}}\
						{LA_NOT, {T_DATA_TYPE_0}, { "declaration__optional", {\ {LA_IS, {""}, 0, { "" }}\ }}}\
value = [sign] , unsigned_value;	value = sign__optional, unsigned_value;	value → sign__optional unsigned_value	value(1: "0" , "1" , "2" , "3" , "4" , "5" , "6" , "7" , "8" , "9" , " + " , " _ ") → sign__optional unsigned_value	value = -sign >> unsigned_value >> BOUNDARIES;	value = sign__optional >> unsigned_value >> BOUNDARIES;	{LA_IS, { "unsigned_value_terminal" , T_ADD_0 , T_SUB_0 }, { "value", {\ {LA_IS, {""}, 2, { "sign__optional", "unsigned_value" }}\ }}}\
	sign__optional = sign ε;	sign__optional → sign sign__optional → ε	sign__optional(1: "+" , " _ .") → sign sign__optional(1: ! "+" , ! " _ .") → ε		sign__optional = sign "";	{LA_IS, {T_ADD_0 , T_SUB_0 }, { "sign__optional", {\ {LA_IS, {""}, 1, { "sign" }}\ }}}\
						{LA_NOT, {T_ADD_0 , T_SUB_0 }, { "sign__optional", {\ {LA_IS, {""}, 0, { "" }}\ }}}\
sign = sign_plus sign_minus;	sign = sign_plus sign_minus;	sign → sign_plus sign → sign_minus	sign(1: "+") → sign_plus sign(1: "-") → sign_minus	sign = sign_plus sign_minus;	sign = sign_plus sign_minus;	{LA_IS, {T_ADD_0}, { "sign", {\ {LA_IS, {""}, 1, { "sign_plus" }}\ }}}\
						{LA_IS, {T_SUB_0}, { "sign", {\ {LA_IS, {""}, 1, { "sign_minus" }}\ }}}\
sign_plus = "+" ;	sign_plus = "+" ;	sign_plus → "+"	sign_plus(1: "+") → "+"	sign_plus = SAME_RULE(tokenPLUS);	sign_plus = SAME_RULE(tokenPLUS);	{LA_IS, {T_ADD_0}, { "sign_plus", {\ {LA_IS, {""}, 1, {T_ADD_0}}\ }}}\
sign_minus = "- " ;	sign_minus = "- " ;	sign_minus → "- "	sign_minus(1: " _ .") → "- "	sign_minus = SAME_RULE(tokenMINUS);	sign_minus = SAME_RULE(tokenMINUS);	{LA_IS, {T_SUB_0}, { "sign_minus", {\ {LA_IS, {""}, 1, {T_SUB_0}}\ }}}\

				BOUNDARY__TAB, \ BOUNDARY__VERTICAL_TAB, \ BOUNDARY__FORM_FEED, \ BOUNDARY__CARRIAGE_RETURN, \ BOUNDARY__LINE_FEED, \ BOUNDARY__NULL, \ NO_BOUNDARY	BOUNDARY__SPACE, \ BOUNDARY__TAB, \ BOUNDARY__VERTICAL_TAB, \ BOUNDARY__FORM_FEED, \ BOUNDARY__CARRIAGE_RETURN, \ BOUNDARY__LINE_FEED, \ BOUNDARY__NULL, \ NO_BOUNDARY	
						\ \ \ { LA_IS, { T_NAME_0 }, { "program____part1", { \ { LA_IS, { "" }, 7, { T_NAME_0, "program_name", \ T_SEMICOLON_0, T_BODY_0, T_DATA_0, \ "declaration_optional", T_SEMICOLON_0 }} \ } } } \ } \ } \ "program_rule"